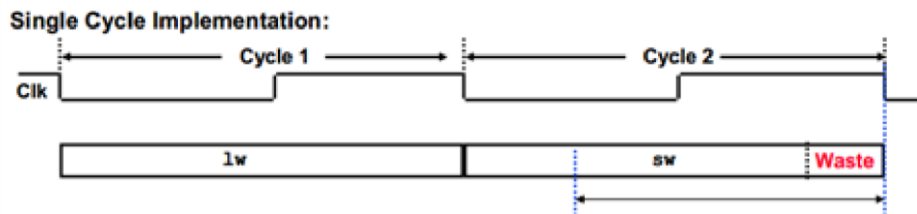


Tutorial 9: Processor, Memory System and Computer Peripherals

Q1. Use suitable diagram to show the timing aspect for single-cycle RISC processor for a minimum of two consecutive clock cycles. List the disadvantages of using single-cycle processor implementation.

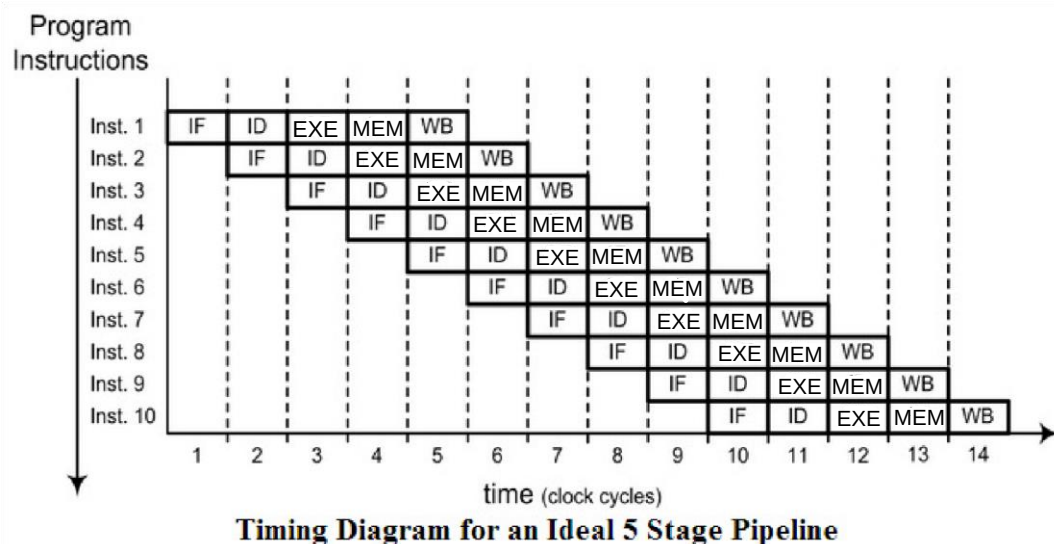


Disadvantages:

- Inefficient use of clock cycle, it is set to accommodate the slowest instruction, and must be the same length for all instructions
 - All instructions take as much time as the slowest instruction than needed
 - It is problematic for complex instructions like floating point multiplication
- Need to duplicate resources (eg. Functional Units) that are used more than once per cycle – waste area and power

Q2. Describe the advantages of conventional RISC processor pipelined execution by using a suitable diagram.

Conventional 5-stages pipelined execution representation as shown below:



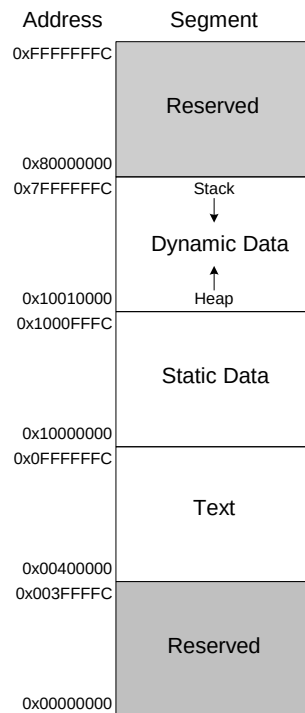
Advantages:

- Enable several instructions operating at once by using the idle datapath resources simultaneously – overlapping instruction stages execution

- Each instruction at a different stage of its execution, using different resources of the datapath.
- Improves performance by increasing throughput (bandwidth) – more instruction per unit time (average instructions per second or per clock)
- Faster execution for a large amount of instructions

Q3. Draw and explain the MIPS memory map. Then, briefly describe the byte-addressable feature of the memory.

MIPS memory map:



- With 32-bit addresses, the MIPS address space spans 2^{32} bytes $\approx$ 4 GB.
- MIPS memory allocation is divided into 3 parts:
 - Text segment
 - Static data segment
 - Dynamic data segment
- The size of each segment can be calculated as follows:

Text segment:

$32'h1000_0000 \sim 32'h0040_0000$

$\Rightarrow 32'h0FC0_0000$ bytes

$\Rightarrow 264,241,152$ bytes (264MB)

$\Rightarrow 66,060,288$ words (66M instructions)

Static data segment:

$32'h1000_FFFF - 32'h1000_0000$

$\Rightarrow 32'hFFFF$ bytes

=> 65,536 bytes (64KB)
=> 16,383 words (16K 32-bit operands (data/address))

Dynamic data segment:

32'h7FFF_FFFC - 32'h1000_FFFF

=> 32'h6FFE_FFFD bytes

=> 1,878,982,653 bytes

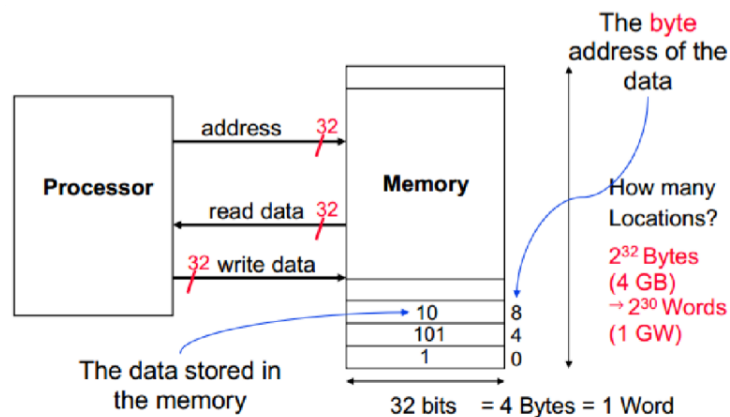
=> 469,745,663 words (469M 32-bit operands (data/address))

Stack:

- Size is not known in advance
- Max size same as heap

- The memory map shown is a virtual memory
 - It is not the actual physical memory (sdram)
 - The size is much larger than the physical memory
 - Physical memory is limited to a portion of the address bits (e.g., 26 bits)
- Memory allocation is not a property of hardware
 - It is a convention used by many MIPS system
 - With the same set of rules, software written by different people can work together effectively on hardware

Byte-addressable feature:



- Memory is a large, single-dimensional array
 - 32-bit address acts as the index into the memory array
- MIPS processor assumed byte granularity
 - The smallest element that the processor can access is a byte
- Memory is organized as byte addressable (each byte has a unique address)
 - Word access (4 bytes) must be aligned
 - Half word and single byte access need not align

Q4.What are the key differences between programmed I/O, interrupt-driven I/O, and DMA, and how do they affect the efficiency and performance of data transfer in computer systems?

Programmed I/O – The CPU initiates the I/O operation, waits for the peripherals to be ready, and transfers data byte by byte. The programmed I/O is simple to implement but inefficient as it requires the CPU to continuously check the status of devices.

Interrupt-Driven I/O – CPU initiates the I/O operation then does other stuff. When the peripheral device is ready, it generates an interrupt signal to the CPU then CPU transfers the data. Interrupt-Driven I/O is more efficient compared to programmed I/O as the CPU can perform other tasks while waiting for the device.

Direct Memory Access (DMA) – CPU asks the DMA controller to transfer data then CPU can continue with other tasks. The DMA controller requests the I/O device, waits and manages the transfer of data between the device and main memory. Once finished, the DMA controller interrupts the CPU. DMA reduces CPU overhead and improves the performance but it requires hardware support (using a DMA controller)

Q5.Explain why is serial interface more popular compare to parallel interface in today's computer system.

Serial interface is popular than parallel interface because:

- Serial interface required lesser data lines to implement.
- Parallel interface is usually plaque by data skew issue.
- Parallel interface is only useful for short distance.

Q6.Explain what is NRZI encoding in USB data transmission and how it helps to prevent the loss of synchronization.

- In NRZI encoding, the presence or absence of a signal transition (change in voltage level) represents the binary data.
- A zero in data being transmitted is represented by a transition from 1 to 0 or 0 to 1.
- A one is represented by a fixed value of 1 or 0.
- To prevent loss of synchronization, a 0 is stuffed in before every seventh bits position in a string of 1s prior to NRZI encoding – maximum six 1s appear in series.